SALES PREDICTION USING PYTHON

Project Report

CodeAlpha Internship — Task 4

A machine learning project exploring how advertising spend relates to product sales.

1. About the Project

For this project, I used Python and machine learning to explore how spending on different advertising platforms can help predict sales. The dataset included advertising spend for TV, radio and newspaper, together with the sales recorded for each observation. I used the advertising figures as the information given to the models and asked them to estimate sales.

2. Project Aim

The aim was to understand whether advertising spend could be used to make useful sales predictions. I also wanted to compare different regression models, see which advertising platforms had the strongest relationship with the model's predictions, and try the model with a new example campaign.

3. Understanding the Dataset

The dataset contained 200 rows and 5 columns: TV, Radio, Newspaper, Sales and an additional unnamed index column. The three advertising columns were used as input features, while **Sales** was the value I wanted to predict. The dataset did not include information such as customer demographics or target segments, so those were not used in this project.

4. Preparing and Exploring the Data

- I loaded the CSV file into a pandas DataFrame and looked at the first few rows to understand its layout.
- I checked the number of rows and columns, the data types and summary statistics.
- I checked for missing values and duplicate records as part of the data-quality review.
- I used scatter plots to explore how TV, radio and newspaper advertising spend related to sales.
- I created a correlation table and heatmap to compare the relationships between the numerical columns.
- I selected TV, Radio and Newspaper as the input features (X), and Sales as the target (y).
- I split the data into training and testing sets, using 80% for training and 20% for testing, with random_state=42.

5. Models I Tried

I trained two regression models. I started with **Linear Regression**, which provides a straightforward way to estimate how the input advertising figures are associated with sales. I then trained a **Random Forest Regressor**, which combines predictions from multiple decision trees and can learn more complex patterns in the data. The Random Forest model used 200 trees and random_state=42.

6. Understanding the Linear Regression Coefficients

I looked at the coefficients from the Linear Regression model to see how each advertising platform contributed to its estimate, while keeping the other inputs constant. The values from my model were:

Advertising platform	Coefficient
TV	0.044730
Radio	0.189195
Newspaper	0.002761

In this fitted Linear Regression model, Radio had the largest coefficient, followed by TV. Newspaper had a much smaller coefficient. These values describe the model's estimated relationship per one unit of advertising spend,

with the other inputs held constant. They should not be treated as proof that advertising on a platform directly causes sales to increase.

7. Random Forest Model Results

I evaluated the Random Forest model using R^2, Mean Absolute Error (MAE) and Mean Squared Error (MSE). R^2 indicates how well the model explains variation in the test-set sales values. MAE shows the average absolute prediction error, while MSE gives greater weight to larger errors.

Evaluation metric	Result
Test R^2 score	0.9818
Test MAE	0.6287
Test MSE	0.5734
Average 5-fold cross-validation R^2	0.9758

The Random Forest achieved an R^2 score of about 0.982 on the held-out test set. Its average five-fold cross-validation R^2 was about 0.976. The closeness of these scores is encouraging and suggests that the model performed consistently across the splits used in this evaluation. However, these results are based on this dataset and evaluation setup, so they do not guarantee the same performance on every future advertising campaign.

8. Trying a New Advertising Campaign

After training the Random Forest model, I tested it with a new example campaign using these advertising spend values: TV = 200, Radio = 30 and Newspaper = 40. The model predicted sales of approximately **18.28** (in the dataset's sales units). This showed how the trained model could be used to generate an estimate for a new set of advertising inputs.

9. What I Learned

Working on this project helped me practise the steps involved in a regression task, from loading and exploring data to visualising relationships, training models and evaluating predictions. I also learned how to compare Linear Regression with Random Forest and how cross-validation can provide another way to check model performance. Looking at the coefficients helped me understand how the fitted Linear Regression model used each advertising input.

10. Conclusion

Overall, this project showed that the advertising figures in the dataset can be used to make sales estimates. The Random Forest model produced strong results on the test set and in five-fold cross-validation. The coefficient analysis also showed that Radio had the largest coefficient in the Linear Regression model, although this should be interpreted as a model relationship rather than a cause-and-effect result. In future, I could test additional data, try further model tuning and include more information about customers or campaigns if it became available.

Tools Used

Python, pandas, NumPy, Matplotlib, Seaborn and scikit-learn.

Note: The results and prediction in this report are based on the outputs recorded during the project. The dataset's sales and advertising units have not been labelled because their exact units were not confirmed.